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Lecture Notes «Matemātiskā analīze I»

CALCULUS I

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LECTURE 5.

DERIVATIVE

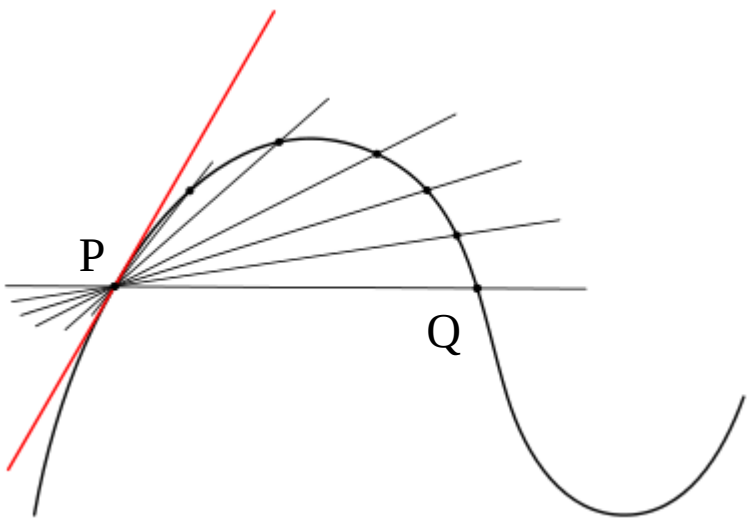
Tangent line
Instantaneous velocity
Definition of the derivative
Derivative and continuous functions
Rules of finding derivatives
Derivatives of the elementary functions
5. Homework

Tangent line

Let P be a arbitrarily chosen point in a curve and let Q be a nearby movable point on that curve.

DEFINITION 5.1. The line that goes through the points P and Q is called secant line.

DEFINITION 5.2. The tangent line at P is the limiting position (if it exists) of the secant line as Q moves toward P along the curve and reaches it.



Tangent line

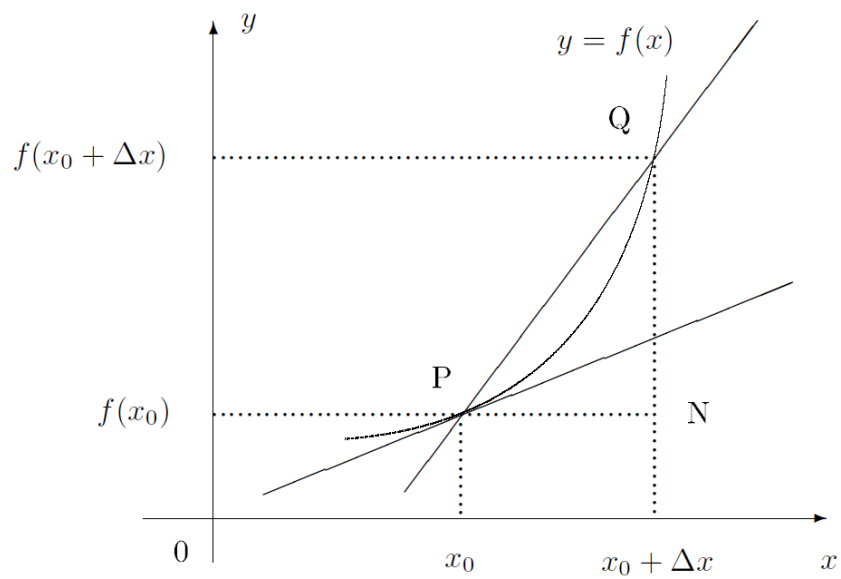
Suppose that the curve is the graph of the equation $y = f(x)$.
Then P has coordinates $(x_0, f(x_0))$,
a nearby point Q has coordinates $(x_0 + \Delta x, f(x_0 + \Delta x))$,
and the secant line through P and Q has slope

$$m_{sec} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \frac{|QN|}{|PN|}$$

The tangent line (if it exists) to the curve $y = f(x)$ at the point $\mathrm{P}(x_0, f(x_0))$ is that line through P with slope

$$m_{tan} = \lim_{\Delta x \to 0} m_{sec} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

To find the equation of the tangent line
$y = m_{tan}x + b$ with the known slope m_{tan}
at the point (x_0, y_0), we should find b.
As we know that the line goes through (x_0, y_0),
then $y_0 = m_{tan}x_0 + b$ or $b = y_0 - m_{tan}x_0$.
Therefore we can write the equation of
the tangent line at the point (x_0, y_0) as
$y - y_0 = m_{tan}(x - x_0)$



Instantaneous velocity

Function $x = x(t)$ that describes the dependency of the coordinates of a material point from time t is called the motion law of the matreial point.

Suppose that a material point moves along a coordinate line so that its position at time *t* is given by $x = x(t)$. At time moment *t* the coordinate of the point is *x(t)*, but a the nearby time $t + \Delta t$, it is $x(t + \Delta t)$.

The average velocity in interval $[t, \Delta t]$ is defined as the distance from the first position to the second position divided by the elapsed time:

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

For describing the velocity at the time moment t, the instantaneous velocity notion is introduced. It is limit of the v_{avg} when $\Delta t \to 0$.
If an objest moves along coordinate line with position function x(t), then its instantaneous velocity at time t is

$$v = \lim_{\Delta t \to 0} v_{avg} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}$$

Definition of the derivative

Velocity is only one of many rates of change. Other rates of chage that could interest us are density for a wire (the rate of change of mass in respect to distace), marginal revenue (the rate of change of revenue with respect to the number of items produced), current (the rate of change of electrical charge with respect to time) etc.
In each case, we must distinguish between an average rate of change on an interval and an instantaneous rate of change at a point.

We have seen that slope of the tangent line and instantaneous velocity are manifestations of the same basic idea.
Rate of growth of an organism (biology), marginal profit (economics), density of a wire (physics, and dissolution rates (chemistry) are other version of the same concept.
Good mathematical sense suggest that we study concept independently of these specialized vocabularies and diverse applications. We choose the neutral name derivative. Add it to function and limit as one of the key words in calculus.

Definition of the derivative

The derivative of a function *f* is another function *f'* (read «f prime») whose value at any numver *x* is

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}.$$

If this limit does exist, we say that *f* is differentiable at *x*.

Finding derivative is called differentiation.

Thus, the derivative of a function f is limit of the ratio between change in function and change in argument when change in argument approaches 0. We can use also notation

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Definition of the derivative

The derivative of the function $f(x) = x^3$ at a point 7 can be given as

$$f'(7) = \lim_{\Delta x \to 0} \frac{(7+\Delta x)^3 - 7^3}{\Delta x} \; or \; f'(7) = \lim_{x \to 7} \frac{x^3 - 7^3}{x - 7}$$

There are several <u>denotions</u> for the derivative:

$$f', y', Df, \frac{dy}{dx}, \frac{df}{dx}$$

Example

If $f(x) = \sqrt{5x-2}, x > \frac{2}{5}$, find $f'(x)$.

$$f'(x) = \lim_{\Delta x \to 0} \frac{\sqrt{5(x+\Delta x)-2} - \sqrt{5x-2}}{\Delta x} = \lim_{\Delta x \to 0} \frac{5(x+\Delta x)-2-(5x-2)}{\Delta x(\sqrt{5(x+\Delta x)-2} + \sqrt{5x-2})} =$$

$$= \lim_{\Delta x \to 0} \frac{5\Delta x}{\Delta x(\sqrt{5(x+\Delta x)-2} + \sqrt{5x-2})} = \frac{5}{2\sqrt{5x-2}}. \blacksquare$$

Exercise

Find the derivate of the function $y = x^2 + 2x$ at a point 5.

Derivative and continuous functions

If a curve has a tangent line at a poit, then that curve cannot take a jump or wiggle too bad at the poit, that is, it is continuous at that point.

Theorem 5.5. If $f'(x)$ exists, then f is continuous at x.

The converse of this theorem is false. If a function f is continuous at x, it does not follow that f has a derivate at x.

It is easy seen by considering $f(x)=|x|$ at the point 0. This function is certainly continuous at zero. However, it oes not have derivative there.

$$\lim_{\Delta x \to 0} \frac{|0+\Delta x| - |0|}{\Delta x} = \lim_{\Delta x \to 0} \frac{|\Delta x|}{\Delta x}, \quad \text{but}$$

$$\lim_{\Delta x \to 0+} \frac{|\Delta x|}{\Delta x} = \lim_{\Delta x \to 0+} \frac{\Delta x}{\Delta x} = 1 \neq -1 = \lim_{\Delta x \to 0-} \frac{-\Delta x}{\Delta x} = \lim_{\Delta x \to 0-} \frac{|\Delta x|}{\Delta x}.$$

Rules of finding derivatives

Theorem 5.6. If *f(x)* and *g(x)* are differentiable functions, then

1. $\left(kf(x)\right)' = kf'(x), k \in \mathbb{R} - constant$ Constant multiple rule
2. $\left(f(x) + g(x)\right)' = f'(x) + g'(x)$ Sum rule
3. $\left(f(x) - g(x)\right)' = f'(x) - g'(x)$ Difference rule
4. $\left(f(x)g(x)\right)' = f'(x)g(x) + f(x)g'(x)$ Product rule
5. $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}, g(x) \neq 0$ Quotient rule

Let's prove 2. rule

$$(f(x) + g(x))' = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) + g(x+\Delta x) - f(x) - g(x)}{\Delta x} =$$

$$= \lim_{\Delta x \to 0} \left(\frac{f(x+\Delta x) - f(x)}{\Delta x} + \frac{g(x+\Delta x) - g(x)}{\Delta x} \right) =$$

We can divide, because separate limits exist:

$$= \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} + \lim_{\Delta x \to 0} \frac{g(x+\Delta x) - g(x)}{\Delta x} = f'(x) + g'(x). \blacksquare$$

Derivatives of the elementary functions

Theorem 5.7. The following formulas are valid:

1. $k' = 0, k \in \mathbb{R} - constant$ Constant function rule
2. $(x^n)' = nx^{n-1}, n \in \mathbb{R}$ Power rule
3. $(sinx)' = cosx$
4. $(cosx)' = -sinx$
5. $(tgx)' = \frac{1}{cos^2 x}$
6. $(ctgx)' = -\frac{1}{sin^2 x}$
7. $(a^x)' = a^x lna, a > 0$
8. $(e^x)' = e^x$
9. $(\log_a x)' = \frac{1}{xlna}, a > 0, a \neq 1$
10. $(lnx)' = \frac{1}{x}$
11. $(arcsinx)' = \frac{1}{\sqrt{1-x^2}}$
12. $(arccosx)' = -\frac{1}{\sqrt{1-x^2}}$
13. $(arctgx)' = \frac{1}{1+x^2}$
14. $(arcctgx)' = -\frac{1}{1+x^2}$

e^x

$lnx = \log_e x$

e - "Euler's number" is a mathematical constant, it is irrational number, approximately equal to 2,718281828459

Derivatives of the elementary functions

Some proofs

1. $(k)' = \lim\limits_{\Delta x \to 0} \frac{k-k}{\Delta x} = \lim\limits_{\Delta x \to 0} \frac{0}{\Delta x} = \lim\limits_{\Delta x \to 0} 0 = 0$

3. $(\sin x)' = \lim\limits_{\Delta x \to 0} \frac{\sin(x+\Delta x)-\sin x}{\Delta x} =$

$= \lim\limits_{\Delta x \to 0} \frac{\sin x \cos \Delta x + \cos x \sin \Delta x - \sin x}{\Delta x} =$

$= \lim\limits_{\Delta x \to 0} \left(-\sin x \frac{1-\cos \Delta x}{\Delta x} + \cos x \frac{\sin \Delta x}{\Delta x}\right) =$

$= -\sin x \lim\limits_{\Delta x \to 0} \frac{1-\cos \Delta x}{\Delta x} + \cos x \lim\limits_{\Delta x \to 0} \frac{\sin \Delta x}{\Delta x} =$

$= -\sin x \cdot 0 + \cos x \cdot 1 = \cos x,$

5. $(\mathrm{tg} x)' = \left(\frac{\sin x}{\cos x}\right)' = \frac{(\sin x)' \cos x - \sin x (\cos x)'}{(\cos x)^2} =$

$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}.$ ■

Derivatives of the elementary functions

Examples

1. $(3x^4 - \sqrt{x} + 5\sin x)' = 12x^3 - \frac{1}{2\sqrt{x}} + 5\cos x.$

2. $((2x^3 - x + 2)\ln x)' = (2x^3 - x + 2)'\ln x + (2x^3 - x + 2)(\ln x)' =$

$$= (6x^2 - 1)\ln x + \frac{2x^3 - x + 2}{x}.$$

3. $$\left(\frac{12 + 3\sin x}{\cos x}\right)' = \frac{(12 + 3\sin x)'\cos x - (12 + 3\sin x)(\cos x)'}{\cos^2 x} =$$

$$= \frac{3\cos x\cos x - (12 + 3\sin x)(-\sin x)}{\cos^2 x} = \frac{3\cos^2 x + 12\sin x + 3\sin^2 x}{\cos^2 x} =$$

$$= \frac{3 + 12\sin x}{\cos^2 x}. \blacksquare$$

Exercises. Please finf *f'(x)* if:

1) $f(x) = \frac{12}{x^6} + \frac{3}{x} - 4\sqrt[3]{x^2} + 7\arcsin x;$

2) $f(x) = (2^x + 3\cdot 5^x)(\operatorname{tg} x - \operatorname{ctg} x).$

HOMEWORK 5

9. Please, find the derivatives by using the definition.

a) $f(x) = \frac{2x-1}{x-4}$

b) $f(x) = \frac{3}{\sqrt{x-2}}$

Answers: a) $\frac{-7}{(x-4)^2}$, b) $\frac{-3}{2(x-2)\sqrt{x-2}}$

10. Please, find the derivatives by using rules and formulas.

a) $f(x) = 3x^{-4} + \frac{5}{\sqrt{x}} - 2\sqrt[5]{x^3} + lnx + 5sinx - 7 * 9^x$

b) $g(t) = (15t^4 - 6t + 9) * e^t$

c) $h(x) = \frac{4x^2-7}{sinx-cosx}$

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p.93 – 118.

Chapter 2. Sections 2.1.-2.4.